



Leather solid waste: An eco-benign raw material for leather chemical preparation – A circular economy example

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ABSTRACT

One of the long lasting problems associated with leather industry is to meet environmental standards for both liquid and solid wastes. Statistics show that one tonne of wet-salted hides/skins yields around 650 kg of solid waste. Among various wastes generated, trimmings for the most part have been underutilized. Collagen presents in trimmings waste are effectively used but hair goes unutilized or at the most as feed for boilers during gelatin manufacturing. Hence, newer technology is needed for complete and effective utilization of raw trimmings. In leather manufacture, formaldehyde condensates polymers are used as re-tanning agent to enhance the compaction of leather. However, these products are hard for biodegradation and also cause the release of free formaldehyde in leather, which is a known carcinogen. Here, there is a need for development of formaldehyde free re-tanning agent for eco-benign leather processing. In this work an attempt had been made to develop formaldehyde free biodegradable eco-benign re-tanning agent from raw trimming of tannery solid waste as a circular economy model. Alkaline (7.5% w/w NaOH) – hydrogen peroxide (10%w/w) pre-treatment followed by thermal hydrolysis at 100 °C for 5 h was an optimized method for effective hydrolysis of trimmings and the process of preparation of product results in the holistic utilization of raw trimmings. The developed product was characterized using Dynamic light scattering and FTIR techniques. The product prepared was further used in leather manufacture as a re-tanning agent and was found to impart multifunctional properties to leathers such as fullness, grain tightness and shade of dye brilliance. Product improves the mechanical strength characteristics of leather and also the exhaustion of post-tanning chemicals. SEM analysis shows that the experimental leather is more compact and flat than control. This novel strategy had not only solved the issue of solid waste but also resulted in a greener leather auxiliary leading to greener environment.

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1. Introduction

Leather industry finds a prominent place in the growth of under developed and developing nations through massive employment generation and foreign exchange (Kumar, 2016). Leather industry utilizes raw hides/skins, by-product of meat industry, as a raw material. Generally, the hides/skins are trimmed prior to leather processing and the trimmings are disposed as solid waste (Kanagaraj et al., 2006). The quantum of trimmings waste generation depends on the raw material source and it's about 5% w/w for hides and 12–15% w/w for skins (Kanagaraj et al., 2015). Statistical data from FAO (Food and Agriculture Organisation) reveal that globally, an average of 418 thousand tons of raw trimmings waste is generated per year (FAO, 2013). Collagen and keratin are the two structural proteins present in hides/skins trimmings where the col-

lagen content is higher than keratin. Collagen present in the raw trimmings is utilized for several industrial applications such as preparation of industrial grade gelatin (Smith and Donovan, 1982), polypeptides (Olle et al., 2013) and adhesives (Liu Wentao et al., 2011). CN Patent-1102358830A deals with preparation of protein based adhesive from raw trimmings such a way that enzymatically hydrolysing the collagen followed by crosslinking with glyoxal. Recently, Masilamani et al. (2016), attempted to extract collagen from raw trimmings using propionic acid/acetic acid and they reported that the propionic acid extraction (93%) showed higher yield than acetic acid (85%) extraction. Attempt was also been made to develop a biodegradable packing material from tannery solid waste (Masilamani et al., 2017). The hydrolyzed collagen was used as an additive in textile (Paul et al., 2012) and leather dyeing (Fang Ren et al., 2011) to improve the dye uptake. Thus it is evident that the collagen present in raw trimmings waste was effectively utilized but remaining keratin (hair) is being disposed as secondary waste. The disposed hair waste is predominantly

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